



GREENHOUSE GAS MONITORING PLAN

Eastern Dunes Field Central Processing Facility 1 (CPF-1)

Document number	DEC-EDF-CPF1-HSE-MP-001
Revision / date	Rev 4.0 · 20 March 2025
Status	Submitted to EAD
Reporting period covered	From 1 January 2025 until superseded
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Revision history

Rev	Date	Description
1.0	30-May-2024	Initial Monitoring Plan following EAD registration
2.0	12-Sep-2024	Quantified methodology added for all methane sources (section 6)
3.0	05-Dec-2024	VRU-401 availability monitoring and LDAR quantification method added
4.0	20-Mar-2025	Annual update for RY2025

1. Purpose and regulatory basis

This Monitoring Plan sets out how Dunes Energy Company monitors and reports direct (Scope 1) emissions of carbon dioxide and methane from Eastern Dunes Field Central Processing Facility 1 (CPF-1). It implements Article 6 of Federal Decree-Law No. 11 of 2024 on the Reduction of Climate Change Effects and the Agency's Technical Guidance for MRV of Greenhouse Gas Emissions in Abu Dhabi Emirate, under the operational control approach.

2. Facility description and boundary

Eastern Dunes Field Central Processing Facility 1 (CPF-1) is an onshore oil processing facility in the Eastern Dunes Field, Al Dhafra Region, operated by Dunes Energy Company since 2009. Well fluids are separated in HP and LP inlet separators. Crude is stabilised and stored in two fixed-roof tanks (T-401A/B) whose vapours are recovered by VRU-401 before export by pipeline. Associated gas is compressed by two gas-turbine-driven compressors (K-201A/B), dehydrated in a TEG unit (U-250) and exported as sales gas or reinjected. Power is generated on site by three gas turbine generators. Fuel gas is taken from the dehydrated gas stream. An HP and an LP flare handle relief, purge and upset gas. Diesel is used only for the emergency generator and fire water pumps. The facility boundary covers all equipment within the CPF fence; wells and flowlines are included, the export pipeline beyond the custody transfer meter is excluded (operated by a third party).

Tag	Unit	Capacity	Source stream	Notes
GTG-A/B/C	Gas turbine generators	3 x 24 MW	SS-01 Fuel gas	
K-201A/B	Gas compressors, gas-turbine driven	2 x 11 MW	SS-01 Fuel gas	Wet seals
H-101A/B	Crude heaters (direct fired)	2 x 18 MW(th)	SS-01 Fuel gas	
FL-501	HP flare	Design 4.2 MMSm3/d	SS-02 HP flare gas	Continuous pilot

Tag	Unit	Capacity	Source stream	Notes
FL-502	LP flare	Design 0.6 MMSm3/d	SS-03 LP flare gas	Continuous pilot gas
EDG-1 / P-801A/B	Emergency diesel generator, fire water pumps	2.5 MW / 2 x 0.8 MW	SS-04 Diesel	Standby
U-250	TEG dehydration unit	3.0 MMSm3/d	-	Still vent (methane source)
T-401A/B	Crude storage tanks	2 x 250,000 bbl	-	Vapours to VRU-401
VRU-401	Vapour recovery unit (electric)	0.25 MMSm3/d	-	Installed 2023

3. Site schematic

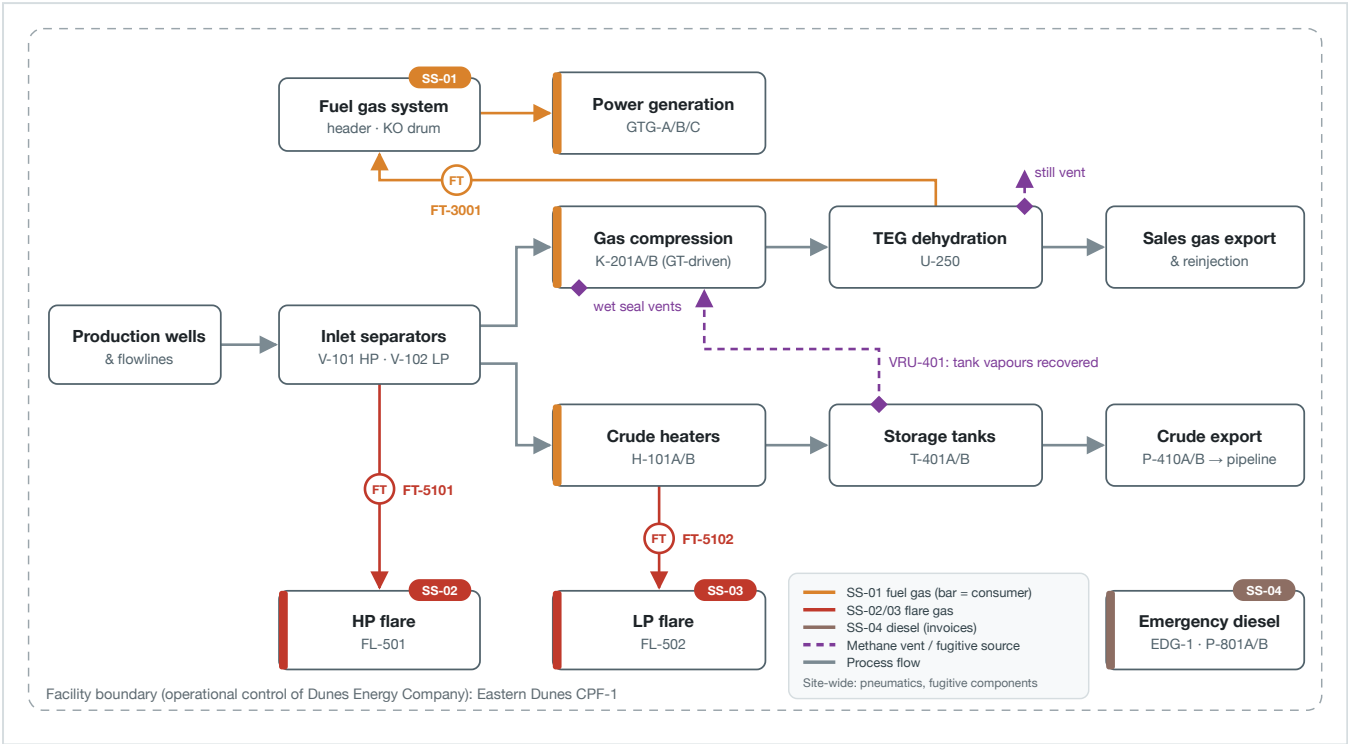


Figure 1: Emission sources, source streams and measurement points at Eastern Dunes CPF-1. Tags are referenced throughout this plan and the emissions report.

4. Source streams and categorisation

Estimated emissions are based on the reporting year 2024. Categories follow the Technical Guidance (Appendix 1.3, Step 3).

ID	Source stream	Emission sources	Est. t CO2 (2024)	Category	Tier
SS-01	Fuel gas	ES-01 GTG-A/B/C, ES-02 K-201A/B drivers, ES-03 H-101A/B	158,141	Major	Tier 3
SS-02	HP flare gas	ES-04 HP flare FL-501	61,126	Major	Tier 2
SS-03	LP flare gas	ES-05 LP flare FL-502	6,792	Minor	Tier 2
SS-04	Diesel	ES-06 EDG-1, ES-07 P-801A/B	3,441	De-minimis	Tier 1

5. Calculation methodology

5.1 Combustion of fuel gas and diesel

$$CO_2 \text{ (t)} = AD \text{ (Sm}^3\text{)} \times NCV \text{ (MJ/Sm}^3\text{)} \times 10^{-6} \times EF \text{ (t CO}_2\text{/TJ)} \times OF \quad | \quad \text{Diesel: } CO_2 \text{ (t)} = m \text{ (t)} \times NCV \text{ (GJ/t)} / 1000 \times EF \text{ (t CO}_2\text{/TJ)}$$

5.2 Calculation factors

Parameter	Source stream	Basis	Tier
NCV fuel gas	SS-01	Site-specific: quarterly GC analysis at SP-3001 (ISO 6974 / ISO 6976), arithmetic mean of four samples	Tier 3
EF fuel gas	SS-01	Site-specific: calculated from the carbon content of the same quarterly samples	Tier 3
Oxidation factor	SS-01, SS-04	1.0	-
EF flare gas	SS-02, SS-03	Composition-based from quarterly GC at SP-5101, combustion efficiency 0.98	Tier 3
NCV / EF diesel	SS-04	IPCC 2006 defaults (43 GJ/t, 74.1 t CO ₂ /TJ)	Tier 1

5.3 Flaring

$$CO_2 \text{ (t)} = V \text{ (Sm}^3\text{)} \times [\sum (x_i \times n_i) \times CE + x_{CO_2}] \times 44.01 / 23.645 / 1000$$

x_i = mole fraction of hydrocarbon component i , n_i = carbon atoms per molecule, CE = combustion efficiency, standard conditions 15 °C and 101.325 kPa.

6. Methane monitoring

All methane sources at the facility are quantified as follows. Results are reported in sheet G of the emissions report.

ID	Source	Method	Data / frequency
M-01	Flare combustion slip	Metered flare volume $\times x_{CH_4} \times (1 - 0.98)$	FT-5101/5102 daily; GC quarterly
M-02/03	Fuel gas and diesel combustion	IPCC 2006 Tier 1 (1 and 3 kg CH ₄ /TJ)	Monthly energy use
M-04	Storage tanks T-401A/B	Process simulation of flash + working/breathing losses $\times (1 - \text{VRU availability})$	Simulation annual; VRU run-hours monthly
M-05	TEG still vent (U-250)	GRI-GLYCalc v4	Operating data quarterly
M-06	Gas-driven pneumatic controllers	Device inventory $\times$ device-specific bleed factor	Inventory annual; bleed survey 2024
M-07	Compressor wet seals K-201A/B	Measured degassing vent flow $\times$ CH ₄ content	Quarterly measurement
M-08	Fugitive components	OGI LDAR survey; Hi-Flow quantification of leaks	Quarterly survey (ClearSight OGI Surveys LLC)
M-09	Maintenance blowdowns	Vented volume (P, V, T of isolated section) $\times$ CH ₄ content	Per event (permit-to-work)

VRU-401 availability target 97%; pneumatic inventory 42 gas-driven devices.

7. Measurement equipment

Tag	Service	Type	Range	Tier	Calibration interval	Calibrated by
FT-3001	Fuel gas header	Ultrasonic, 4-path, DN300	0-15,000 Sm ³ /h	Tier 3	12 months	Gulfmetric Calibration Services LLC
FT-5101	HP flare header	Ultrasonic flare gas meter, 2-path, DN600	0.03-100 m/s	Tier 2	12 months	Gulfmetric Calibration Services LLC
FT-5102	LP flare header	Ultrasonic flare gas meter, 2-path, DN300	0.03-100 m/s	Tier 2	12 months	Gulfmetric Calibration Services LLC

Gas sampling points: SP-3001 (fuel gas header) and SP-5101 (HP flare header), sampled quarterly and analysed by Sabkha Analytical Laboratories LLC.

8. Data flow, QA/QC and data gaps

8.1 Data flow

Meter totals are recorded in the PI historian, extracted monthly into the GHG workbook (DEC-EDF-CPF1-HSE-PR-110), reviewed by the HSE Manager, compiled into the annual emissions report and verified by an accredited verifier before submission to the Agency.

8.2 Control activities

- Monthly reconciliation of FT-3001 fuel gas totals with the production allocation fuel gas figure (tolerance 2%).
- Monthly reconciliation of flare meter totals against the gas balance in the production allocation report. Deviations greater than 10% are investigated and documented.
- Diesel deliveries reconciled with supplier invoices and tank dips.
- Year-on-year comparison of emissions and intensity; variances greater than 10% explained in the annual review memo.

8.3 Data gaps and substitution

Where a meter is unavailable, missing data are substituted with the average of the 30 days of valid data preceding the gap. Gaps are recorded in sheet H1 of the emissions report and notified to the Agency within 30 days. Where a meter is expected to be out of service for more than 30 days, this Monitoring Plan is revised and resubmitted to the Agency within 30 days of the change.

8.4 Revision of this plan

The plan is reviewed annually and updated before 1 April each year, and within 30 days of any significant change in operations, methodology or measurement equipment.

9. Roles and responsibilities

Post	Responsibility
Facility Manager	Accountable for MRV compliance; approves this plan and signs the operator declaration.
GHG & Energy Lead	Owens this plan, the GHG workbook and the annual report; runs monthly reconciliations.
Instrumentation Supervisor	Calibration and maintenance of FT-3001, FT-5101 and FT-5102.
Laboratory Coordinator	Quarterly gas sampling and liaison with the analytical laboratory.
Production Accounting Engineer	Monthly hydrocarbon allocation and gas balance.

10. Records

All monitoring data, calculations, calibration certificates, analyses, invoices and verification reports are retained for at least five years.

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20 March 2025

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20 March 2025

